ERASTUS MURUNGI

SOFTWARE ENGINEERING

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Clubs

MIT African Student Association · Secretary — Sept. 2021 to Current

National Society of Black Engineers (NSBE) — Sept. 2019 to Current

Skills

CODE

C/C++
Python
Java
HackLang
TypeScript
Go
Matlab
iOS development
Android Development
Compiler development

Education

Massachusetts Institute of Technology — Sept. 2019 to Current
B.S. Electrical Engineering and Computer Science 2023
Relevant Coursework: Performance Engineering, Computer Language Engineering, Computer Systems Engineering, Foundations of Computer Security, Foundations of Statistics
Current GPA: 4.0/5.0

Software Engineering Professional Experience

Meta — Menlo Park
Software Engineering (SWE) Intern — May 2022 to Aug. 2022

- Increased the number of volunteering signups by 17% by developing a messaging tab in Facebook Community Help across Web, Android, and iOS, to enable volunteers and organizers to connect.
- Increased the number of volunteering opportunities views by 3% by proposing and developing a new "See more" link on the volunteering details page.
- Collaborated with privacy control experts to modularize privacy control policies in the Community Help team at Meta. Surveyed engineers reported improved usability and higher satisfaction while using the *Privacy Rules* API.
- Skills used: **HackLang, ReactJS, Android Development, iOS Development, Native Templates, Screenshot Testing**

Facebook — Menlo Park
Software Engineering (SWE) Intern — June 2021 to Aug. 2021

- Awarded *Better Engineering* honor by Business Suite VP at Meta for proposing and implementing the migration of the Business Suite Android codebase to a new dependency injection framework.
- Increased Business Suite Android app's message inbox views by 6.7% by implementing and deploying the inbox suggestions control settings feature on the Business Suite Android to help business managers filter inbox reply suggestions and by collaborating with two other engineers to ensure uniformity of the feature across iOS and the web.
- Improved null safety in a codebase of more than 20,000 lines of Android code from Java to Kotlin.
- Skills used: **Android Development, Dogfooding, Logging, A-B Testing, GraphQL**

Facebook — Cambridge, MA
Facebook University for Engineering Intern — June 2020 to Aug. 2020

- Collaborated with peers to develop 4 mobile apps, publicly available on GitHub.
- Developed and thoroughly tested a fully functioning android mobile app with mentoring from 3 Facebook Android engineers.
- Skills used: **Java, Kotlin**

Leadership and Activities

Machine Learning · Lab Assistant — Sept. 2022 to Current

- Managed lab sessions and conducted check-ins to ensure students have a thorough understanding of the material.
- Conducted weekly office hours for 8 hours each week in blocks to help students apply concepts in machine learning to theory and lab assignments.

MIT Digital Humanities Lab: Self-Sufficient Cities · Research Assistant — Feb. 2022 to May 2022

- Collaborated with Professor Kate Brown to develop an online, interactive mapping and timeline project that tells the story of the rise and fall of urban communities that grew their own food in the 20th century United States.
- Used machine learning to collect and aggregate data used in the project from population and agricultural census documents, sample household survey data, and satellite imagery.

Digital Communication Systems:Lead Lab Assistant — Sept. 2020 to Dec. 2021

- Helped students for 10 hours each week in applying concepts of digital communication systems such as error-correcting codes, signal processing, and networking to their lab assignments and achieve correct and performant results in class labs and quizzes.
- Collaborated with Professor Katrina LaCurts to design and review labs done by students in the course.

MIT Digital Humanities Lab: Gender Analysis Toolkit · Research Assistant — Sept. 2021 to Dec. 2021

- Led a team of 6 to use natural language processing to analyze the use of gender in over 4200 novels sourced from the Gutenberg project.
- Used Cython and parallelization to optimize memory and runtime of the python code written because of the large corpus of over 4200 novels and used d3-js and ReactJS to visualize the results of the analysis.
- Presented the project at the end of the semester to a panel of students and professors and made the tool available online at http://gendernovels.digitalhumanitiesmit.org/.

Projects

FastMalloc — Nov. 2021 to Nov. 2021

- As a way of applying learnings from MIT's Performance Engineering class (6.172), I designed and implemented a fast and thread-safe dynamic memory allocator, which ranked first in a performance derby of projects at the end of the semester.

Orient — July 2020 to Aug. 2020

- With the goal of streamlining the process of orientation for new students joining MIT, I built and tested an Android app for institutions to send students critical information and for students to connect during orientations. The app follows the MVVM software architectural pattern, has chatroom functionality, and utilizes Google Maps SDK to embed locations of orientation events.
- Presented the app to a panel of more than 50 Facebook engineers.

Decaf, an SSA-based optimizing compiler — Sept. 2021 to Current

- Due to the scarcity of implementations of readable source code for static single assignment (SSA) based compilers, I built a compiler for a C-like language called Decaf using SSA techniques. Designed and developed a strongly-typed SSA-based intermediate representation, similar to that of the industry-strength LLVM compiler, and implemented LLVM-style compiler optimization passes i.e sparse conditional constant propagation.
- Generated assembly code for most small programs on macOS matched that produced by LLVM.

IPMJulia, an Interior Point Method based solver — May 2021 to June 2021

- To satisfy my intellectual curiosity and to provide more open source material on solving optimization problems in the Julia language, I implemented a linear program problem solver using the interior point method algorithm.
- Solutions produced match those produced by the commonly used COIN-OR Branch-and-Cut solver.